

# A Multi-Source Binaural Sound Scene Using HRIR-Based Spatial Rendering

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**Abstract**—This paper presents a 30-second binaural sound scene combining multiple static and dynamic sources in a coherent three-dimensional auditory environment. Spatial rendering is achieved through HRIR convolution: static sources via direct convolution, moving sources via block-based time-varying processing. To mitigate perceptual discontinuities caused by the discrete spatial sampling of the HRIR dataset, temporal smoothing is applied between successive impulse responses, transforming abrupt transitions into continuous motion.

Time-varying audio effects are processed prior to spatialisation, ensuring that transformations are perceived as originating from specific locations within the scene. Listening tests confirm stable externalisation and reliable azimuthal tracking, while revealing expected limitations in vertical localisation. Separating static and dynamic rendering paths improves computational efficiency. The work demonstrates that perceptually grounded design choices enable the construction of convincing multi-source binaural scenes within the constraints of discretely sampled spatial data.

**Index Terms**—Binaural audio, HRIR, spatial sound, time-varying convolution, virtual acoustics.

## I. INTRODUCTION

This project presents a 30-second binaural radio drama that explores the construction of a three-dimensional sound scene using head-related impulse responses. The listener is placed at the centre of a small interior space while multiple sound sources appear and move around them, allowing spatial events to be perceived in both azimuth and elevation.

The scene unfolds as follows: a door opens behind the listener, an elderly male voice greets them and walks in a circle while speaking, accompanied by footsteps. A vintage radio is switched on to the left side, and the scene concludes with a moving voice crossing overhead from left to right above the listener's head. These elements demonstrate static positioning, horizontal and vertical motion, and the integration of audio effects such as distortion, delay, and flanging.

The creative and technical aims of the demonstration are to: (1) construct a coherent multi-source binaural soundscape, (2) combine static and moving sound sources within the same scene, (3) explore elevation cues in addition to horizontal-plane motion, and (4) integrate time-varying audio effects within the spatial rendering pipeline. Spatial positioning is achieved using head-related impulse responses (HRIRs) from the SADIE II database (subject H19). Static sources remain fixed in space, while moving sources follow predefined trajectories around the listener.

Sound material was prepared from multiple sources. Dialogue lines were generated using ElevenLabs text-to-speech technology, while environmental sounds, such as door openings, footsteps, and room tone, were sourced from the Soundly sound effects library. The final dialogue events were recorded in an anechoic chamber to ensure fully dry material for dynamic spatialisation and modulation. All material was pre-edited, organised, and temporally aligned in PreSonus Studio One 7 before being imported into MATLAB for binaural processing.

Section II details the theoretical context and implementation of the spatial rendering engine and effects pipeline. Section III evaluates the system's performance, discussing both perceptual outcomes and design iterations.

## II. CONTEXT AND IMPLEMENTATION

### A. Binaural Cues and the Binaural Field

Human sound localisation relies primarily on interaural time differences (ITD) at low frequencies and interaural level differences (ILD) at high frequencies due to head shadowing, but these binaural cues alone produce spatial ambiguities such as the cone of confusion [1]–[3]. Elevation and front-back discrimination are instead resolved through direction-dependent spectral filtering introduced by the pinna [1].

As illustrated in Figure 1, these cues vary systematically with azimuth for SADIE II subject H19: ITD follows an azimuth-dependent trend reflecting path-length differences, while ILD increases at higher frequencies. Ear-input spectra further exhibit direction-dependent shaping associated with the outer ear, supporting elevation and front-back perception [1].

These binaural cues are encoded directly in the impulse responses. As shown in Figure 2, the offset between left- and right-ear onsets reflects ITD, while the magnitude responses exhibit direction-dependent colouration from pinna and head filtering. Together, these properties constitute the physical substrate imposed on source signals by HRIR convolution.

### B. HRIR-Based Binaural Rendering

Head-Related Transfer Functions (HRTFs) describe the transfer function between a localised free-field anechoic source and the signals present at a listener's tympanic membrane [4]. Typical features include a direction-dependent time delay, spectral colouration caused by the pinnae, and contributions

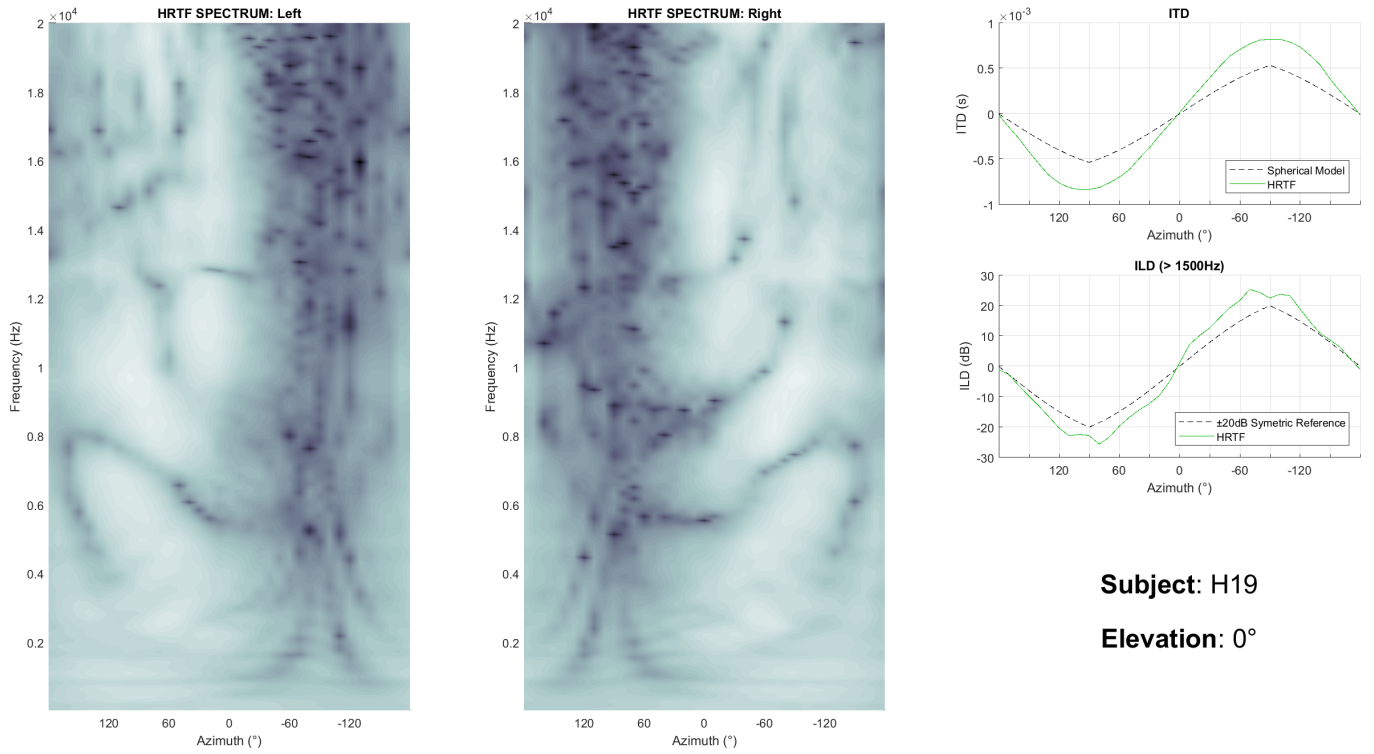


Fig. 1. Binaural cues for SADIE II subject H19 at  $0^\circ$  elevation. Left: magnitude spectra of the left-ear HRTF as a function of azimuth and frequency. Centre: magnitude spectra of the right-ear HRTF. Right: interaural time difference (ITD) and interaural level difference (ILD) as functions of azimuth, compared with idealised spherical-head references. The plots illustrate the systematic azimuth-dependent variation of ITD and ILD, as well as the direction-dependent spectral shaping introduced by the outer ear. Adapted from [6]

from the shoulders and torso. In the time domain, these functions correspond to head-related impulse responses (HRIRs), which provide a practical representation for signal processing.

Alternatively, Binaural Room Impulse Responses (BRIRs) incorporate additional environmental components such as wall and floor reflections [5]. In this work, anechoic HRIRs are used rather than BRIRs so that all spatial cues arise from listener-related filtering alone. The HRIRs are drawn from the SADIE II database, which provides high-resolution, diffuse-field equalised measurements over dense spherical grids for human listeners and dummy heads [6].

A binaural signal is obtained by convolving a monophonic source with the HRIR pair associated with a desired spatial direction [6]. For a source located at azimuth  $\theta$  and elevation  $\phi$ , the rendering process can be expressed as

$$y_L(n) = x(n) * h_L(\theta, \phi), \quad y_R(n) = x(n) * h_R(\theta, \phi), \quad (1)$$

where  $x(n)$  is the input signal and  $h_L(\theta, \phi)$  and  $h_R(\theta, \phi)$  are the left and right HRIRs for that direction.

While this formulation is mathematically compact, it conceals the rich temporal and spectral structure of the HRIRs themselves. Figure 2 makes these properties explicit for a representative direction by showing both the time-domain impulse responses and their corresponding magnitude spectra.

In practice, the SADIE II database samples the binaural field at a finite set of azimuth–elevation positions rather

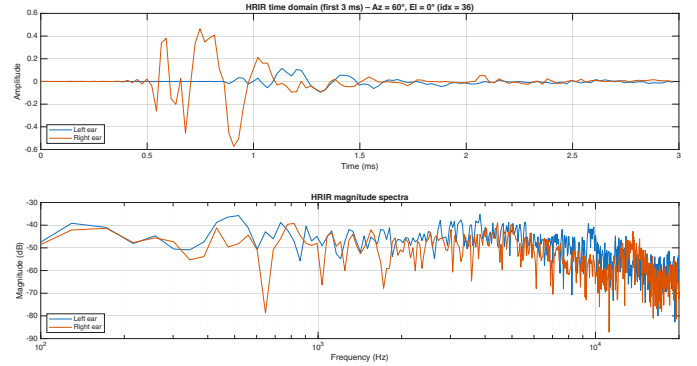


Fig. 2. Time-domain HRIR pair and spectra for SADIE II subject H19 (azimuth  $60^\circ$ , elevation  $0^\circ$ ), showing ITD and direction-dependent HRTF colouration.

than providing a continuous representation [6]. For subject H19, this corresponds to a grid of 50 measured directions, imposing a quantised spatial resolution. Spatial rendering therefore consists of selecting the HRIR pair that best matches a desired source position and applying the convolution in (1). In this system, this mapping is performed by the function `closestHRIR`, which computes the spherical distance between a target direction and all available measurement points using the haversine (3) metric and returns the index of the nearest HRIR.

### C. Convolution-Based Rendering: Static and Dynamic Sources

Once a spatial direction has been mapped to an HRIR pair, binaural rendering reduces to convolution. For static sources, the HRIR remains constant over time and the binaural signal can be obtained by a single direct convolution of the input with the selected HRIR pair. This corresponds to linear time-invariant (LTI) filtering, where a fixed impulse response models the spatial transfer function [7]. The `static_renderer` function handles this operation by performing a one-shot convolution and produces a fixed spatial image for the entire duration of the sound.

For moving sources, the spatial filter becomes time-varying, as the HRIR must change with source position. A single global convolution is therefore no longer valid. Instead, the signal is processed in short, overlapping frames, each rendered with the HRIR corresponding to the current source direction. This is realised in `dynamic_renderer` using a block-based Overlap-Add (OLA) scheme [7], in which the input is segmented into overlapping frames, each windowed and convolved in the frequency domain with the appropriate HRIR, and recombined by overlap and addition in the time domain.

Early experiments revealed that source motion sounded perceptually stepped, reflecting the finite spatial resolution of the HRIR grid. As the source crossed between adjacent measurement directions, abrupt HRIR changes produced audible discontinuities. To mitigate this effect, successive HRIRs are smoothed over time using a first-order recursive crossfade:

$$\mathbf{h}_n = (1 - \alpha) \mathbf{h}_{n-1} + \alpha \mathbf{h}_n^*, \quad (2)$$

where  $\mathbf{h}_n^*$  is the HRIR corresponding to the target direction at frame  $n$ ,  $\mathbf{h}_{n-1}$  is the previously smoothed HRIR, and  $\alpha \in (0, 1)$  controls the transition rate.

The choice of the smoothing coefficient  $\alpha$  represents a trade-off between temporal responsiveness and spectral stability. A value close to 1 results in fast transitions but risks preserving the audible steps between HRIRs, while a value close to 0 produces very smooth transitions but introduces a perceptual lag. Through empirical testing, a coefficient of  $\alpha \approx 0.1$  (for a block size of 512 samples at 44.1 kHz) was found to provide the optimal balance. This value ensures that the spectral interpolation occurs rapidly enough to track the source's angular velocity without introducing audible phasing artefacts.

### D. Trajectory Control and HRIR Selection

In this system, source motion is defined in terms of azimuth and elevation trajectories over time. Azimuth denotes the horizontal angle around the listener's head in the transverse plane, while elevation specifies the vertical angle relative to that plane [1]. A moving source is therefore described by an initial direction  $(az_{ini}, el_{ini})$ , a final direction  $(az_{end}, el_{end})$ , and a temporal window  $[t_{start}, t_{end}]$ .

The function `trajectory` converts this specification into a frame-wise angular path by linearly interpolating azimuth and elevation between the start and end points across the corresponding time interval. This produces continuous angular

motion in the perceptual domain, independent of the discrete sampling of the underlying HRIR dataset.

Because the SADIE II dataset provides HRIRs only at a finite set of measured directions [6], each continuous angular position generated by `trajectory` must be mapped onto a discrete HRIR index. This mapping is performed by `closestHRIR`, which selects the measurement direction that best matches the target azimuth-elevation pair. Rather than relying on simple angular differences, this selection is based on spherical distance on the unit sphere, ensuring that proximity is evaluated in a geometrically meaningful way for directional data [1].

The distance between a target direction  $(az, el)$  and each measured direction  $(az_i, el_i)$  is computed using the haversine formulation on the unit sphere,

$$d_i = 2 \arcsin \sqrt{\sin^2 \left( \frac{el_i - el}{2} \right) + \cos(el) \cos(el_i) \sin^2 \left( \frac{az_i - az}{2} \right)}, \quad (3)$$

and the index corresponding to the minimum distance is returned.

As a result, each frame of the dynamic renderer is associated with a single HRIR index, forming the vector `FrameAngle`. This vector constitutes the discrete spatial control signal for the rendering engine: at every processing block, `dynamic_renderer` queries the HRIR specified by `FrameAngle(n)` and updates the spatial filter accordingly. A complete summary of the scene's sound objects, defining their trajectories and associated signal processing chains, is provided in Table I.

TABLE I  
SUMMARY OF SOUND OBJECTS, SPATIAL BEHAVIOUR, AND PROCESSING IN THE BINAURAL SCENE.

| Source       | Type    | Trajectory (Az/El)     | Effect                 |
|--------------|---------|------------------------|------------------------|
| Room Tone L  | Static  | 315°, 0°               | –                      |
| Room Tone R  | Static  | 45°, 0°                | –                      |
| Room Tone C  | Static  | 0°, 0°                 | –                      |
| Room Tone Ls | Static  | 225°, 0°               | –                      |
| Room Tone Rs | Static  | 135°, 0°               | –                      |
| Door         | Dynamic | 130° → 80°, 45° → –45° | –                      |
| Footsteps 1  | Dynamic | 110° → 310°, –45°      | –                      |
| Dx1          | Dynamic | 110° → 310°, 30°       | –                      |
| Radio        | Static  | 310°, –20°             | Dist., Delay, HPF, LPF |
| Music        | Static  | 310°, –20°             | Dist., Delay, HPF, LPF |
| Footsteps 2  | Dynamic | 310° → 450°, –45°      | –                      |
| Dx2          | Dynamic | 340° → 450°, 30°       | –                      |
| Dx3          | Dynamic | 270°, –45° → 45°       | Flanger                |
| Dx4          | Dynamic | 90°, 45° → –45°        | Flanger                |

### E. Integration of Audio Effects in the Spatial Pipeline

While trajectory algorithms determine the spatial location of a source, the perceptual identity of the sound objects is established through signal processing. In this demonstration, audio effects are applied prior to binaural rendering. Each sound source is first processed in mono by its associated effect chain and only then convolved with the corresponding HRIRs. This design ensures that the spectral and temporal modifications introduced by each effect become part of the spatialised signal itself.

The "radio" source is processed with nonlinear distortion followed by a slap-back delay and band-limiting. The distortion effect implements a symmetrical soft-clipping algorithm adapted from Dutilleux and Zölzer [7]. Unlike hard clipping, which introduces aliasing through abrupt truncation, this implementation uses an exponential transfer function to gradually saturate the signal. For an input  $x(n)$  with gain  $G$ , the intermediate drive signal  $q(n) = G \cdot x(n)$  is mapped to the output  $y(n)$  via:

$$f(q) = \text{sgn}(q) \cdot \left(1 - e^{-|q|}\right). \quad (4)$$

The final output is a linear mix of this saturated signal and the original input. This function introduces odd-order harmonics dependent on the input amplitude, effectively simulating the non-linear response of an overdriven analogue circuit.

Following distortion, the signal is processed by a slap-back delay. This is implemented via convolution with a generated Finite Impulse Response (FIR) filter rather than a recursive delay line. The filter impulse response  $h(n)$  consists of a direct path and a series of discrete echoes spaced by the delay time  $\tau$ :

$$h(n) = \delta(n) + \sum_{k=1}^K g^k (-1)^{k-1} \delta(n - k \cdot \tau f_s), \quad (5)$$

where  $g$  is the feedback attenuation coefficient and  $f_s$  is the sampling rate. Notably, the term  $(-1)^{k-1}$  introduces an alternating polarity inversion for successive echoes, creating a specific phase cancellation pattern that colours the decay of the radio source. The cumulative spectral impact of these processing stages is illustrated in Fig. 3.

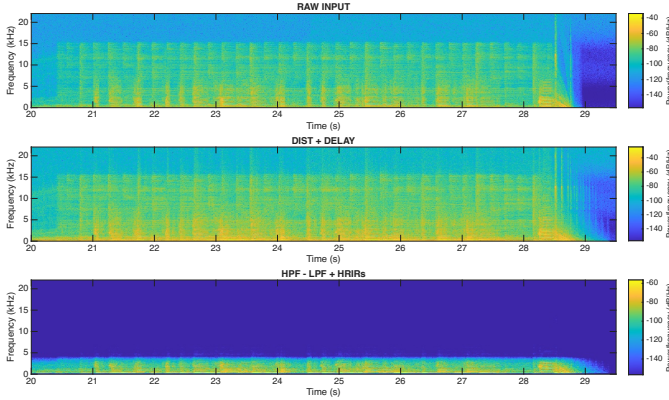


Fig. 3. Spectrogram comparison of the radio source processing stages. Top: raw input signal. Middle: signal after distortion and delay, showing increased harmonic content and repeated echoes. Bottom: signal after band-limiting (HPF+LPF) and HRIR convolution, where the spectrum is confined to a narrow band characteristic of a small loudspeaker.

Similarly, the flanger effect applied to the final moving voice utilises a feedback delay network modulated by a Low-Frequency Oscillator (LFO), as depicted in Fig. 4. The instantaneous delay length  $M(n)$  in samples varies sinusoidally according to:

$$M(n) = M_0 (1 + A \sin(2\pi f_{\text{LFO}} n / f_s)), \quad (6)$$

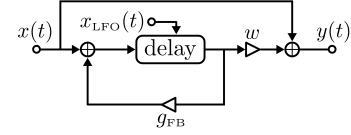


Fig. 4. Block diagram of a flanger effect. A short, LFO-modulated delay (0–15 ms) is mixed with the direct signal, and a feedback path with gain  $g_{\text{FB}}$  produces a time-varying comb-filter response characteristic of flanging. Adapted from [8].

where  $M_0$  is the nominal delay and  $A$  determines the modulation depth. The system implements a recursive difference equation:

$$y(n) = x(n) + w(n - M(n)), \quad (7)$$

where the internal state  $w(n)$  includes a feedback path  $w(n) = x(n) + g_{\text{FB}} \cdot w(n - M(n))$ . This feedback recirculation creates a series of resonant peaks in the frequency response (comb filtering) that sweep across the spectrum as  $M(n)$  oscillates, producing the characteristic “jet plan motion” effect.

From an implementation perspective, this structure is reflected directly in the main control script: effect functions operate on monophonic signals, and their outputs are then passed to either `static_renderer` or `dynamic_renderer`. The complete signal flow is shown in Figure 5.

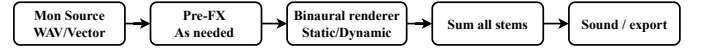


Fig. 5. Block diagram of the complete spatial-audio pipeline. Each sound object is processed in mono by its dedicated effect chain, then passed to either the static or dynamic binaural renderer. All sources are spatialised independently and summed only after HRIR-based rendering, before final playback or export.

## F. Visual Diagnostics and System Verification

The spatial rendering system operates on abstract representations not directly observable in the audio domain, including angular trajectories, HRIR index sequences, and time-varying spatial filters. To support design and verification, visualization tools were developed to project source paths onto a unit sphere surrounding a head model [9]. As illustrated in Figure 6, these diagnostics confirm that continuous angular motion is achieved despite discrete spatial resolution and proved essential during development for identifying discontinuities in angular mapping. Additional views are provided in the Appendix.

## III. EVALUATION

### A. Monitoring and Playback Conditions

The perceptual evaluation of the system was conducted using Audio-Technica ATH-M50x circumaural closed-back headphones. Unlike open-back models, these headphones were selected for their high acoustic isolation, which minimises the interference of external environmental noise during the assessment of subtle spatial cues. Additionally, their detailed transient response facilitates the precise tracking of moving sources and the evaluation of temporal artefacts.

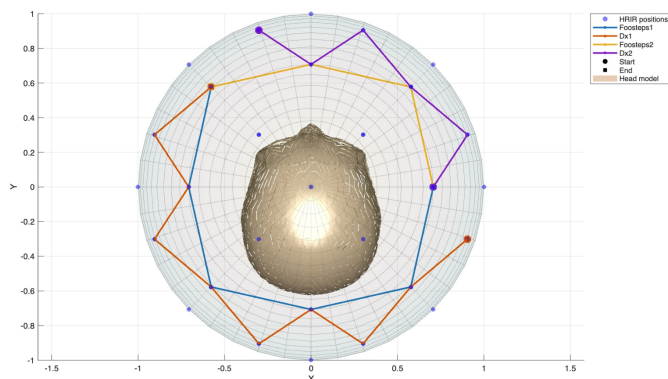


Fig. 6. Top-down view of the realised trajectories of the footsteps and dialogue sources over the complete HRIR measurement field. The discrete HRIR positions define a quantised spatial domain, while the coloured paths show the frame-wise positions selected by the dynamic renderer. The figure illustrates how continuous angular motion is realised within a discretely sampled binaural space.

It is important to note that the SADIE II HRIRs used in this project are diffuse-field equalised. Consequently, no specific headphone compensation filter (HpTF) was applied during monitoring. While individualised headphone compensation can theoretically improve localisation accuracy, the use of industry-standard monitoring headphones provided a familiar reference point.

### B. Perceptual Outcome and Spatial Coherence

Listening tests indicate that the demonstration achieves a stable sense of externalisation, with sound sources perceived as originating outside the listener’s head rather than collapsing to an internalised stereo image, which is a core requirement of effective binaural reproduction [1]. Static elements remain anchored in space, while moving sources can be tracked continuously around the listener, indicating that azimuthal motion is conveyed reliably. The coexistence of static and dynamic objects does not produce spatial masking or ambiguity: individual sources remain perceptually separable even when temporally overlapping.

Elevation cues prove to be significantly less robust than azimuthal localisation. While horizontal motion is perceived clearly and continuously in most cases, vertical displacement, particularly for negative elevations corresponding to sources located below the listener, is weak and often ambiguous. In these regions, motion is perceived more as a timbral change than as a clear spatial shift. This limitation is consistent with known properties of human spatial hearing, in which elevation relies primarily on subtle pinna-induced spectral cues rather than on strong interaural differences [1]. Despite this, transitions across the median plane remain stable, suggesting that front-back discrimination is largely preserved, but that vertical localisation remains the principal perceptual limitation of the system.

### C. Technical Performance and System Behaviour

From a technical perspective, the system operates reliably across the full duration of the demonstration. Both renderers produce stable output without artefacts such as clicks, dropouts, or buffer discontinuities. The separation between `static_renderer` and `dynamic_renderer` proves essential in practice: static sources are rendered through a single linear convolution, while moving sources are processed using block-based time-varying convolution. This division significantly reduces computational load compared to an architecture in which all sources are treated as dynamic, and enables the scene to be rendered efficiently.

The dynamic renderer behaves consistently under time-varying trajectories. The `Overlap-Add` scheme ensures continuity between frames, while the recursive HRIR smoothing prevents abrupt spectral changes when the source crosses between adjacent measurement directions. Without this smoothing stage, early prototypes exhibited perceptually stepped motion caused by the discrete nature of the HRIR grid. The introduction of the first-order crossfade transforms these discontinuities into gradual spectral transitions, yielding perceptually continuous motion even though the underlying dataset remains quantised.

### D. Reflection and Design Insights

The development of the system was strongly iterative, with both the technical architecture and compositional approach evolving in response to perceptual and computational constraints. Early prototypes revealed two central issues: excessive computational cost when all sources were treated as dynamic, and perceptually stepped motion caused by abrupt transitions between discretely sampled HRIRs.

These observations motivated key design adaptations. To address the stepped motion, temporal smoothing was introduced between successive HRIRs. While this increased the computational load of the dynamic renderer, it yielded a substantial perceptual benefit by transforming abrupt spectral changes into continuous spatial motion. To offset this cost, the processing architecture was split into distinct static and dynamic paths, ensuring that immobile sources avoid the overhead of frame-based convolution.

This iterative process also influenced compositional decisions. It became clear that complex spatial paths were more effective when decomposed into consecutive segments. For instance, the overhead arc was implemented as two separate movements, rising to the apex and then descending to ensure the trajectory remained stable and strictly defined, preventing the spatial image from drifting unintentionally during the transition.

Similarly, early experiments dictated the strict signal flow in which audio effects are applied prior to spatialisation, ensuring that modulation and distortion remain spatially coherent.

Finally, the need to bridge the gap between abstract parameters and perceptual outcome motivated the development of the trajectory visualisation tool. Taken together, these technical and creative refinements enabled the construction of a coherent

multi-source scene that effectively integrates static positioning, vertical motion, and time-varying effects, fulfilling the project's aims within the boundaries of HRIR-based rendering.

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# APPENDIX A SUPPLEMENTARY FIGURES

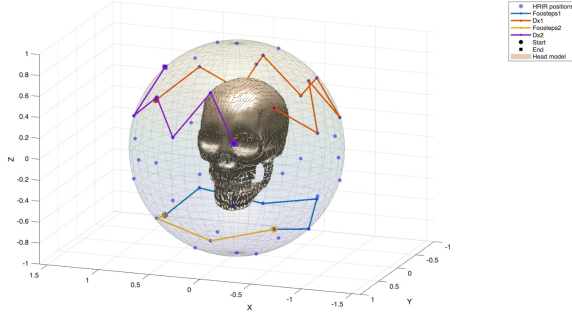


Fig. 7. Front-diagonal view of the movement trajectories of the footsteps and dialogue sources.

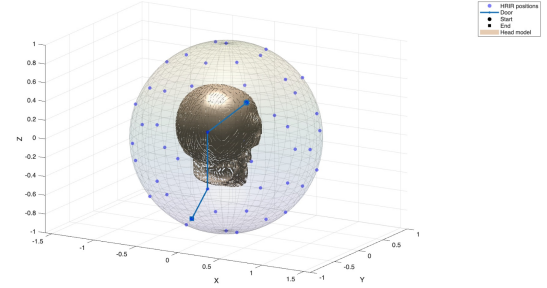


Fig. 8. Back-diagonal view of the movement trajectories of the Door source.

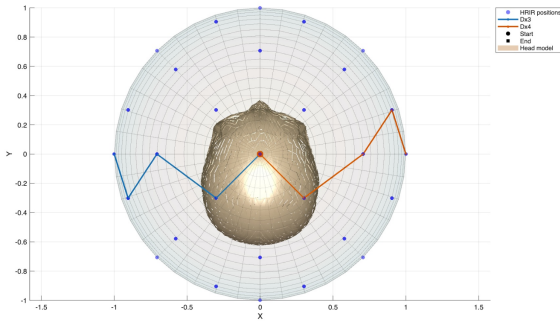


Fig. 9. Top view of the movement trajectories of the final dialogue sources.

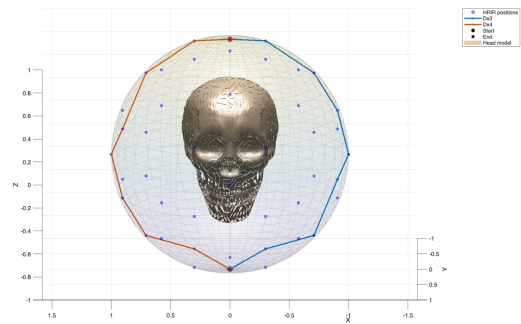


Fig. 10. Front view of the movement trajectories of the dialogue source.

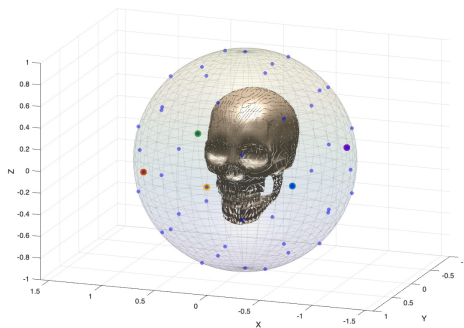


Fig. 11. Front-diagonal view of the static trajectories of the room-tone sources.

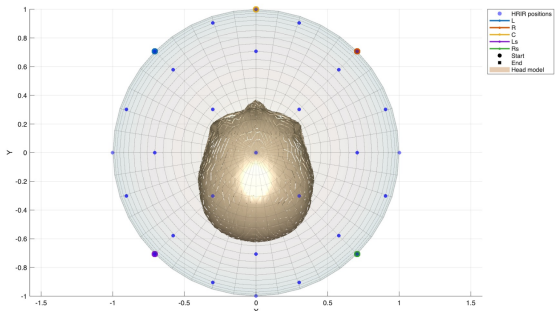


Fig. 12. Top view of the static trajectories of the room-tone sources.

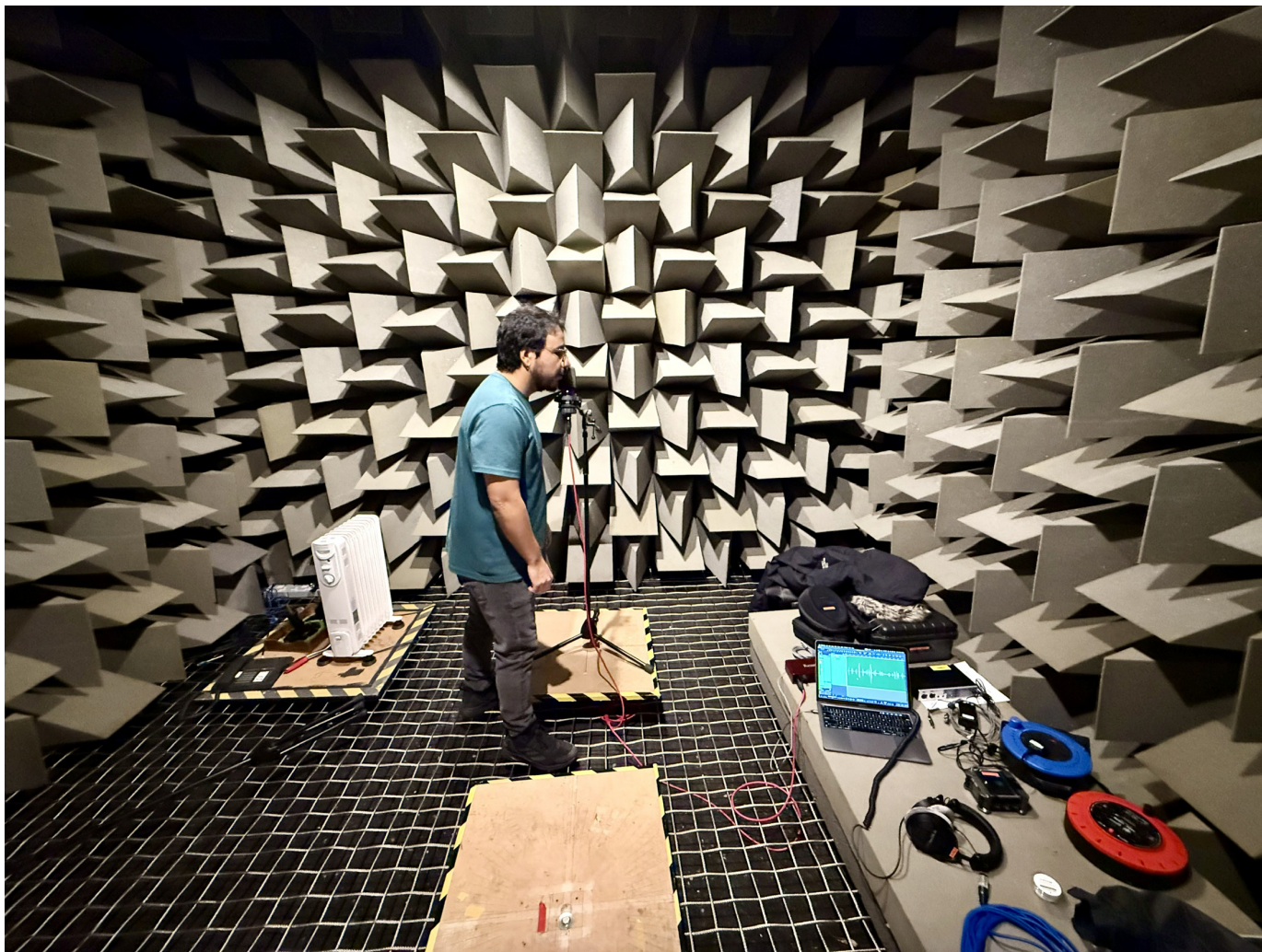


Fig. 13. Dialogue recording in an anechoic chamber, used for the final moving voice processed with flanging